

INCOME CLASSIFICATION AND CUSTOMER SEGMENTATION REPORT

Contents

1	Introduction	2
2	Dataset Description and Structure	2
2.1	Target Definition and Sample Weighting	2
2.2	Feature Overview	3
2.3	Dataset Characteristics	3
3	Data Exploration and Preprocessing	3
3.1	Key Data Insights	4
3.2	Data Quality Issues	4
3.3	Preprocessing Pipeline	4
4	Income Classification Model	4
4.1	Problem Definition	5
4.2	Models Used	5
4.3	Training Strategy	5
4.4	Evaluation Metrics	5
4.5	Results Summary	5
4.6	Model Interpretability and Key Drivers of Income	6
5	Segmentation Analysis	7
5.1	Overview	7
5.2	Feature Selection	7
5.3	Dimensionality Reduction with UMAP	7
5.4	Clustering with HDBSCAN	7
5.5	Cluster Analysis	8
6	Limitations and Ethical Considerations	9
6.1	Temporal Limitations and Concept Drift	9
6.2	Sampling Bias and Representativeness	9
6.3	Feature Limitations	9
6.4	Fairness and Ethical Considerations	9
6.5	Limitations of the Segmentation Pipeline	9
6.6	Summary	10
7	Conclusion	10

1 Introduction

This project examines income prediction and population segmentation using data from the U.S. Census Current Population Survey. The analysis addresses two complementary objectives: predicting whether an individual is a high-income earner and identifying meaningful subgroups within the population based on demographic, employment, and socioeconomic characteristics.

The first objective is formulated as a supervised classification problem. The goal is to predict whether an individual's income exceeds \$50,000 using demographic, employment, and socioeconomic features. In addition to predictive performance, emphasis is placed on model interpretability to identify the factors most strongly associated with income level.

The second objective is formulated as an unsupervised segmentation problem. Rather than predicting a predefined outcome, this analysis seeks to uncover natural groupings within the population. The resulting segments provide a descriptive view of population structure and can support the development of targeted marketing strategies, customer profiles, and resource allocation approaches.

Together, these approaches provide both individual-level predictions and group-level insights. Classification supports the identification of higher-income individuals, while segmentation provides a broader understanding of population heterogeneity. Combined, they offer a data-driven foundation for marketing and customer analysis.

2 Dataset Description and Structure

The dataset used in this project is derived from the 1994–1995 Current Population Survey conducted by the U.S. Census Bureau. It contains 199,523 observations and 40 features describing demographic, employment, socioeconomic, and household characteristics. The dataset also includes a binary income label and a sample weight variable used to account for survey design.

2.1 Target Definition and Sample Weighting

The `label` variable defines a binary income classification task. Individuals are assigned to one of two categories based on annual income:

- `"- 50000."` indicates income less than or equal to \$50,000
- `"50000+."` indicates income greater than \$50,000

The labels are stored as strings and include formatting artifacts such as a trailing period.

The dataset includes a `weight` variable that reflects the survey design and adjusts for unequal sampling probabilities. These weights ensure that evaluation reflects the underlying population distribution rather than the raw sample. In this project, different weighting strategies are explored to address class imbalance, and survey weights are applied during evaluation to produce population-representative performance metrics.

2.2 Feature Overview

The feature set spans several domains and includes both numerical and categorical features:

- **Demographics Attributes:** age, race, sex, hispanic origin, citizenship, country of birth self, country of birth father, country of birth mother
- **Education and Schooling:** education, enroll in edu inst last wk
- **Employment Type and Job Structure:** class of worker, full or part time employment stat, member of a labor union, reason for unemployment, own business or self employed, num persons worked for employer, detailed occupation recode, major occupation code, detailed industry recode, major industry code
- **Income and Financial Indicators:** wage per hour, capital gains, capital losses, dividends from stocks
- **Work History:** weeks worked in year
- **Veteran and Military Status:** fill inc questionnaire for veteran's admin, veterans benefits
- **Household and Family Structure:** marital stat, detailed household and family stat, detailed household summary in household, family members under 18, tax filer stat
- **Geography and Migration History:** region of previous residence, state of previous residence, migration code-change in msa, migration code-change in reg, migration code-move within reg, lived in this house 1 year ago, migration previous residence in sunbelt
- **Temporal Identifiers:** year
- **Additional Features:** weight, label

Many categorical features are encoded as integer codes representing discrete categories, particularly in occupation and industry-related fields. As a result, careful preprocessing is required before modeling.

2.3 Dataset Characteristics

The dataset presents several challenges typical of large-scale survey data. It contains a mix of categorical and numerical features requiring heterogeneous preprocessing strategies. The target variable is highly imbalanced, with a small proportion of high-income individuals. In addition, the data is survey-weighted, requiring careful handling to ensure representative evaluation. The feature space is moderately high-dimensional and contains redundancy across related variables, particularly within occupation, industry, and geographic features. These characteristics motivate both feature selection and dimensionality reduction in later stages of the analysis.

3 Data Exploration and Preprocessing

Prior to model development, the dataset was analyzed to understand its structure, identify data quality issues, and define appropriate preprocessing steps. Given the survey-based nature of the data, particular attention was paid to class imbalance, categorical encoding, missing patterns, and feature relationships.

3.1 Key Data Insights

An initial exploratory analysis was conducted to understand the distributional properties of the dataset and relationships between variables. The target variable is highly imbalanced, with the majority class representing approximately 94% of the data. This imbalance is consistent under both weighted and unweighted views, indicating that it reflects the underlying population structure rather than a sampling artifact. Correlation analysis of numerical variables shows generally weak linear relationships, both among features and with respect to the target. This suggests that income classification is not driven by simple linear effects among numeric attributes. In contrast, categorical variables exhibit stronger structural dependencies. In particular, occupation, industry, education, and household-related features show clear internal relationships and hierarchical structure. Migration-related variables also show evidence of temporal variation between 1994 and 1995, suggesting potential distribution shift in parts of the dataset. Overall, these findings indicate that predictive signal is primarily contained in categorical structure rather than linear relationships in numeric features.

3.2 Data Quality Issues

Several data quality considerations were identified during exploration. Duplicate analysis shows that duplicates are rare and likely reflect survey or processing artifacts. A small number of cases also show identical features with conflicting labels, indicating either noise or limitations in feature resolution.

Missing data is limited but non-uniform. True missing values appear only in the `hispanic origin` feature, while several other variables contain `?` values representing unknown or unrecorded responses. These were treated as missing during preprocessing. The dataset also contains a large number of “Not in Universe” (NIU) values, which indicate structural non-applicability due to survey design rather than missing responses. These values were retained as valid categories, as they carry meaningful information about respondent eligibility.

3.3 Preprocessing Pipeline

Preprocessing was applied consistently across both classification and segmentation tasks, with task-specific feature subsets selected where appropriate. Categorical variables were encoded using one-hot encoding. Missing and unknown values were excluded from encoding to avoid introducing artificial structure. NIU categories were also excluded from one-hot encoding, as their absence is implicitly captured in the remaining feature representation. Numerical variables were standardized using z-score normalization to ensure comparable scaling across features. Although no missing numerical values were present, the pipeline was designed to support median imputation where necessary.

4 Income Classification Model

The classification component of this project focuses on predicting whether an individual falls into a low-income or high-income category based on demographic, employment, and socioeconomic features. This is formulated as a supervised learning problem aimed at supporting targeted marketing and resource allocation.

4.1 Problem Definition

The objective is to predict whether an individual’s income exceeds \$50,000 using structured survey features. In addition to predictive accuracy, interpretability is an important consideration, as the model is intended to provide insight into the factors associated with income differences.

4.2 Models Used

Several supervised learning models were evaluated to balance predictive performance and interpretability:

- **Logistic Regression (LASSO and Ridge):** Linear models used as interpretable baselines, with regularization to handle multicollinearity and high-dimensional feature spaces.
- **Decision Tree (Gini criterion):** A nonlinear model that captures feature interactions and provides rule-based interpretability.
- **XGBoost:** A gradient boosting method selected for its strong performance on structured tabular data and ability to model nonlinear relationships.

4.3 Training Strategy

Model performance was estimated using 5-fold stratified cross-validation to preserve class proportions across folds. Each model was trained under three weighting strategies:

- Original Sample Weights
- Class-Balanced Weights
- Fully Uniform Class Weights

These strategies were used to evaluate the impact of class imbalance handling on model performance. However, all reported evaluation metrics are computed using the original survey-weighted distribution to ensure comparability.

4.4 Evaluation Metrics

Due to the strong class imbalance, multiple evaluation metrics are used rather than accuracy alone. ROC-AUC is used as the primary measure of discriminative ability, while F1 score, precision, recall, specificity, and negative predictive value are used to capture different aspects of classification performance.

4.5 Results Summary

Table 1 presents classification performance for all models under the three weighting strategies.

Across all weighting schemes, XGBoost consistently achieves the strongest overall discriminative performance, particularly in terms of ROC-AUC and F1 score. Logistic regression models behave very similarly to each other and provide competitive performance across all metrics, while the decision tree model generally underperforms, especially in ROC-AUC and F1 score.

Differences between weighting strategies are primarily reflected in threshold-dependent metrics rather than ranking performance. ROC-AUC remains highly stable across all models and weighting schemes, indicating that class reweighting does not materially affect the ability of the models to rank observations by predicted risk. In contrast, class-balanced and uniform weighting consistently increase recall while substantially reducing precision and F1 score across all models, reflecting a systematic shift toward more positive classifications. Overall, the results indicate that all models separate the classes effectively in terms of ranking ability, but differ mainly in how they trade off false positives and false negatives rather than in their underlying discriminative performance.

Table 1: Classification Performance Across Weighting Strategies (mean $\pm$ standard deviation)

Model	ROC-AUC	F1	Precision	Recall
Original Sample Weights				
LASSO Logistic Regression	0.9428 $\pm$ 0.0022	0.4998 $\pm$ 0.0085	0.8379 $\pm$ 0.0053	0.6867 $\pm$ 0.0041
Ridge Logistic Regression	0.9428 $\pm$ 0.0022	0.5015 $\pm$ 0.0086	0.8368 $\pm$ 0.0058	0.6879 $\pm$ 0.0041
Decision Tree	0.7163 $\pm$ 0.0035	0.4646 $\pm$ 0.0058	0.7117 $\pm$ 0.0051	0.7160 $\pm$ 0.0032
XGBoost	0.9510 $\pm$ 0.0018	0.5504 $\pm$ 0.0143	0.8587 $\pm$ 0.0076	0.7118 $\pm$ 0.0071
Class-Balanced Weights				
LASSO Logistic Regression	0.9436 $\pm$ 0.0022	0.4384 $\pm$ 0.0056	0.6410 $\pm$ 0.0024	0.8722 $\pm$ 0.0038
Ridge Logistic Regression	0.9435 $\pm$ 0.0021	0.4388 $\pm$ 0.0052	0.6412 $\pm$ 0.0022	0.8723 $\pm$ 0.0036
Decision Tree	0.7073 $\pm$ 0.0026	0.4457 $\pm$ 0.0050	0.7002 $\pm$ 0.0042	0.7070 $\pm$ 0.0026
XGBoost	0.9506 $\pm$ 0.0018	0.4580 $\pm$ 0.0072	0.6498 $\pm$ 0.0032	0.8785 $\pm$ 0.0041
Uniform Class Weights				
LASSO Logistic Regression	0.9438 $\pm$ 0.0022	0.4357 $\pm$ 0.0046	0.6397 $\pm$ 0.0020	0.8724 $\pm$ 0.0034
Ridge Logistic Regression	0.9437 $\pm$ 0.0020	0.4363 $\pm$ 0.0047	0.6400 $\pm$ 0.0020	0.8725 $\pm$ 0.0033
Decision Tree	0.7098 $\pm$ 0.0054	0.4495 $\pm$ 0.0097	0.7018 $\pm$ 0.0061	0.7094 $\pm$ 0.0054
XGBoost	0.9507 $\pm$ 0.0017	0.4556 $\pm$ 0.0067	0.6486 $\pm$ 0.0030	0.8794 $\pm$ 0.0032

4.6 Model Interpretability and Key Drivers of Income

Interpretability analysis was conducted using logistic regression coefficients, decision tree feature importance, and SHAP values for XGBoost. Despite differences in modeling approaches, a consistent set of predictive drivers emerges across all models. Higher education levels (particularly graduate and professional degrees), increased work intensity (weeks worked per year), and high-status occupations are consistently associated with higher income. In contrast, lower education levels, non-filing tax status, and being female are associated with lower predicted income. Financial variables such as capital gains, dividends, and capital losses also play a significant role, particularly in tree-based models. LASSO and Ridge logistic regression highlight education and occupation as the strongest predictors. The decision tree places greater emphasis on financial variables, while XGBoost captures a broader combination of demographic, financial, and occupational effects through nonlinear interactions. Overall, the consistency across models suggests a stable underlying structure in the data, where income is primarily driven by education, labor market engagement, and financial activity, with additional nonlinear effects captured by more flexible models.

5 Segmentation Analysis

5.1 Overview

The segmentation approach is designed to identify clusters without requiring a fixed number of groups, allow cluster sizes to vary naturally, capture non-linear and irregular cluster structures, and treat ambiguous observations as noise rather than forcing assignment.

The segmentation component of this project is an exploratory, unsupervised analysis designed to identify meaningful subgroups within the population. Unlike the classification task, there are no predefined labels, and the objective is to uncover structure directly from the data based on similarity across demographic, economic, and socioeconomic characteristics. The goal is to produce interpretable segments that reflect real differences in population structure and can support downstream applications such as customer profiling, targeted marketing, and resource prioritization. The segmentation approach is designed to (i) identify clusters without requiring a fixed number of groups, (ii) allow cluster sizes to vary naturally, (iii) capture non-linear and irregular cluster structures, and (iv) treat ambiguous observations as noise rather than forcing assignment.

5.2 Feature Selection

A reduced feature set was selected to improve interpretability, reduce noise in the clustering process, and promote clustering based on meaningful structural differences rather than redundancy or noise in the full feature space. The selected variables capture three complementary dimensions of the population.

- **Demographic Features:** age, sex, race, marital stat, country of birth self, family members under 18
- **Economic Features:** wage per hour, capital gains, capital losses, dividends from stocks
- **Socioeconomic Features:** education, class of worker, major occupation code, major industry code

5.3 Dimensionality Reduction with UMAP

The selected feature space is high-dimensional and sparse due to one-hot encoding of categorical variables. To address this, Uniform Manifold Approximation and Projection (UMAP) is used to embed the data into a lower-dimensional space suitable for clustering. UMAP preserves local neighborhood structure, ensuring that individuals who are similar in the original feature space remain close in the embedded space. This is particularly important in sparse settings where traditional distance metrics become less informative. Cosine distance is used as the similarity metric, as it better captures relationships in sparse, high-dimensional one-hot encoded data by focusing on shared active features rather than absolute magnitude.

5.4 Clustering with HDBSCAN

Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) is applied to the UMAP embedding to identify dense regions corresponding to population segments. HDBSCAN is well suited for this task because it (i) does not require specifying the number of clusters in advance, (ii) allows

clusters to vary in size and shape, and (iii) identifies noise points that do not belong to any cluster. This flexibility is important for socioeconomic data, where population structure is unlikely to follow simple geometric assumptions.

5.5 Cluster Analysis

The clustering procedure identified 28 clusters, which were interpreted post hoc by examining distributions of the original features within each group. Because HDBSCAN does not impose a predefined cluster structure, interpretation relies on differences in demographic composition, employment status, occupation, and financial variables across clusters.

One group of clusters includes noise and highly homogeneous edge cases. Cluster -1 is labeled as noise and contains a large proportion of children and other observations that do not fit well into any dense region of the feature space. Cluster 0 is a degenerate case consisting entirely of 10-year-old male individuals with no labor force participation, reflecting an artifact of extreme feature homogeneity rather than a meaningful socioeconomic segment.

A second group consists of youth and early workforce segments. Clusters 1 and 2 are primarily composed of teenagers in secondary education with early employment experience, often in retail occupations. These clusters differ mainly in gender composition and minor outliers, with Cluster 1 being almost entirely female and Cluster 2 predominantly male.

Several clusters reflect clear occupational and industrial specialization. These include agricultural workers concentrated in farming, forestry, and fishing; government employees separated into federal, state, and local levels; private-sector sales and retail workers; and self-employed contractors concentrated in construction and repair-related services. These clusters are primarily distinguished by occupation and industry codes rather than demographic variables.

Another set of clusters is defined by socioeconomic status and financial characteristics. This includes an investment-oriented group characterized by elevated capital gains and managerial occupations, a professional and technical segment with higher education levels and specialized occupations, and an executive and managerial cluster that is nearly homogeneous in occupational composition.

Finally, a number of clusters are driven primarily by labor market status. These include retired individuals with older age distributions and limited labor participation, as well as unemployed or inactive groups with varying demographic compositions. Additional clusters capture manual and production labor, including manufacturing workers, machine operators, and transportation-related occupations.

Overall, the segmentation reveals a structured population organized primarily along occupation, employment status, education, and financial behavior, with additional stratification by life stage and labor market participation.

The segmentation results reveal a structured population with clear differences in occupation, employment status, education, and financial behavior. The combination of UMAP and HDBSCAN successfully captures both high-level socioeconomic stratification and more granular occupational subgroups, while also identifying noise and edge cases that do not conform to dominant patterns.

6 Limitations and Ethical Considerations

This section outlines key limitations of the data, modeling approaches, and segmentation pipeline, as well as important ethical considerations associated with predictive modeling in socioeconomic contexts.

6.1 Temporal Limitations and Concept Drift

The dataset originates from 1994–1995, which limits the extent to which findings generalize to present-day populations. Relationships between demographic characteristics, employment conditions, and income are likely to have shifted over time due to structural economic changes, labor market evolution, and policy differences. In addition, several feature definitions and coding schemes have changed since the original data collection period, making direct application to modern datasets non-trivial.

6.2 Sampling Bias and Representativeness

Although survey weights are applied, the dataset may still not fully represent the underlying population. Certain subgroups are underrepresented or aggregated into broad categories, which reduces granularity and may introduce bias. As a result, the findings should be interpreted as broad population-level patterns rather than precise estimates for specific subpopulations.

6.3 Feature Limitations

The dataset is restricted to demographic, employment, and basic socioeconomic variables. It does not include behavioral or transactional information such as purchasing history, financial transactions, or digital activity. This limitation reduces the ability to capture fine-grained differences in lifestyle and consumption behavior, which are often important in applied marketing and recommendation systems. Consequently, both predictive performance and segmentation granularity are constrained by the available feature set.

6.4 Fairness and Ethical Considerations

Income prediction and segmentation models may reinforce existing societal biases present in historical census data. Even when sensitive attributes are excluded, many remaining variables (such as education, occupation, and geography) can act as proxies for protected characteristics, potentially leading to indirect discrimination. In the context of clustering, this risk is particularly relevant because unsupervised methods can reproduce socially meaningful groupings without explicit supervision. As a result, clusters may reflect structural inequalities rather than neutral population structure. This highlights the importance of careful interpretation when using these methods for decision-making or targeting applications.

6.5 Limitations of the Segmentation Pipeline

The segmentation approach combines UMAP for dimensionality reduction with HDBSCAN for clustering. While this pipeline is effective for discovering structure, it has several limitations. In particular, HDBSCAN does not naturally support out-of-sample cluster assignment, meaning that new observations cannot be consistently mapped to existing clusters without additional modeling steps. Furthermore, the structure of the embedding space and resulting clusters may be sensitive to hyperparameter choices in both UMAP and

HDBSCAN. This introduces potential instability in cluster assignments, especially when applied to slightly different datasets or updated data distributions.

6.6 Summary

Overall, while the models and segmentation approach provide meaningful insights into income structure and population heterogeneity, their practical application is limited by temporal relevance, feature availability, and methodological constraints. These limitations highlight the importance of cautious interpretation, particularly in applied or decision-making contexts.

7 Conclusion

This project investigated income prediction and population segmentation using U.S. Census Current Population Survey data. The analysis combined a supervised learning task for predicting high-income individuals with an unsupervised clustering approach for identifying meaningful subgroups within the population.

For the classification task, all models achieved strong predictive performance, with XGBoost performing best overall in terms of ROC-AUC and F1 score. Logistic regression models provided competitive performance while offering greater interpretability. Across models, key predictors of income consistently included education level, occupational category, work intensity, and financial variables such as capital gains and dividends. Lower education levels, reduced labor force participation, and lack of investment income were strongly associated with lower income outcomes.

The segmentation analysis revealed a structured population composed of interpretable clusters reflecting occupation, employment status, socioeconomic status, and life stage. The combination of UMAP for dimensionality reduction and HDBSCAN for clustering successfully identified both broad structural groups and finer occupational subsegments, while also separating noise and edge cases that did not conform to dominant patterns.

Overall, the results demonstrate that income structure in the dataset is highly organized and can be effectively modeled using both supervised and unsupervised techniques. However, the analysis is constrained by temporal limitations, feature availability, and methodological considerations that limit direct real-world deployment. Despite these constraints, the study provides a clear demonstration of how machine learning methods can be used to extract both predictive signals and interpretable population structure from census data.